# Database Development and Design (SOE201)

## Lecture 4a – Review of Relational Algebra

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AIAT

# Learning Outcomes

- Six basic relational algebra operators
  - Select: $\sigma$
  - Project: $\Pi$
  - Union: $\cup$
  - Set difference: -
  - Cartesian product: x
  - Rename: $\rho$
- Additional Operators
  - Set intersection: $\cap$
  - Natural join:
  - Division: $\div$
  - Assignment: $\leftarrow$

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# Relation Schema, Instance and Attributes

- $A_1, A_2, \ldots, A_n$ are *attributes*
- $R = (A_1, A_2, \ldots, A_n)$ is *relation schema*
  - Example: *instructor* = (*ID, name, dept_name, salary*)
- Relation instance *r* defined over schema *R* is denoted by *r*(*R*)
  - Current values in relation are specified by table
- Element *t* of relation *r* is called *tuple* and is represented by *row* in table
- Set of allowed values for each attribute is called *domain* of attribute
  - Attribute values are required to be atomic; that is, indivisible
  - Special value *null* is member of every domain, indicating that value is "unknown"



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# Relation Schema, Instance and Attributes – cont'd

- Attributes: *ID, name, dept_name, salary*
- Schema: *instructor* (*ID, name, dept_name, salary*)
- Tuples: rows in relation

| *ID* | *name* | *dept_name* | *salary* |
|---|---|---|---|
| 22222 | Einstein | Physics | 95000 |
| 12121 | Wu | Finance | 90000 |
| 32343 | El Said | History | 60000 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 98345 | Kim | Elec. Eng. | 80000 |
| 76766 | Crick | Biology | 72000 |
| 10101 | Srinivasan | Comp. Sci. | 65000 |
| 58583 | Califieri | History | 62000 |
| 83821 | Brandt | Comp. Sci. | 92000 |
| 15151 | Mozart | Music | 40000 |
| 33456 | Gold | Physics | 87000 |
| 76543 | Singh | Finance | 80000 |



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# Keys

- Let $K \subseteq R$
- $K$ is superkey of $R$ if values for $K$ are sufficient to identify unique tuple of each possible relation $r(R)$
  - Example: {*ID*} and {ID, name} are both superkeys of *instructor*
- Superkey $K$ is candidate key if $K$ is minimal
  - Example: {*ID*} is candidate key for *Instructor*
- One of candidate keys is selected to be primary key
- Foreign key constraint: value in one relation must appear in another
  - Example: *dept_name* in *instructor* is foreign key referencing *department*



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# Relational Algebra, What and Why?

- Similar to normal algebra (as in 2+3*x-y), except we use relations as values instead of numbers.
- Not used as query language in actual DBMSs. (SQL instead)
- Inner, lower-level operations of relational DBMS are, or are similar to, relational algebra operations. One needs to know about relational algebra to understand query execution and optimisation in relational DBMS.
- Some advanced SQL queries requires explicit relational algebra operators, most commonly, *outer join*.
- Relations are seen as *sets of tuples*, which means that no duplicates are allowed. SQL behaves differently in some cases. Remember the SQL keyword distinct.
- SQL is *declarative*, which means that one tells DBMS *what* she wants, but not *how* it is to be calculated. C++ or Java program is *procedural*, which means that one has to state, step by step, exactly how result should be computed. Relational algebra is (more) procedural than SQL. Actually, relational algebra is *mathematical expressions*.

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# Concepts and Operations from Set Theory

- Relations in relational algebra are seen as sets of tuples, so one can use basic set operations.
  - Set
  - Element
  - No duplicate elements (but: multiset = bag)
  - No order among elements (but: ordered set)
  - Subset
  - Proper subset (with fewer elements)
  - Superset
  - Union
  - Intersection
  - Set difference
  - Cartesian product (cross-product)

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# Relational Algebra Operators

- Six basic operators
  - Select: $\sigma$
  - Project: $\Pi$
  - Union: $\cup$
  - Set difference: -
  - Cartesian product: x
  - rename: $\rho$
- Additional operators
  - Set intersection: $\cap$
  - Natural join:
  - Division: $\div$
  - Assignment: $\leftarrow$

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# Definitions

- Basic expression in relational algebra consists of either one of following:
  - Relation in database
  - Constant relation
- Let $E_1$ and $E_2$ be relational algebra expressions; following are all relational algebra expressions:
  - $E_1 \cup E_2$
  - $E_1 - E_2$
  - $E_1 \times E_2$
  - $\sigma_p(E_1)$, $p$ is predicate on attributes in $E_1$
  - $\Pi_s(E_1)$, $S$ is list consisting of some of attributes in $E_1$
  - $\rho_x(E_1)$, x is new name for $E_1$

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# Select Operator

- Notation: $\sigma_p(r)$
- $p$ is called **selection predicate**
- Defined as:

$$\sigma_p(r) = \{t \mid t \in r \text{ and } p(t)\}$$

Where $p$ is formula in propositional calculus consisting of **terms** connected by : $\wedge$ (**and**), $\vee$ (**or**), $\neg$ (**not**)

Each term is:

<attribute> *op* (<attribute> or <constant>)

where *op* is one of: $=, \neq, >, \geq, <, \leq$

- Example of selection:

$$\sigma_{branch_name="Perryridge"}(account)$$

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# Select Operator – Example

Relation r

| A | B | C | D |
|---|---|---|---|
| $\alpha$ | $\alpha$ | 1 | 7 |
| $\alpha$ | $\beta$ | 5 | 7 |
| $\beta$ | $\beta$ | 12 | 3 |
| $\beta$ | $\beta$ | 23 | 10 |

$\sigma_{A=B \wedge D > 5}(r)$

| A | B | C | D |
|---|---|---|---|
| $\alpha$ | $\alpha$ | 1 | 7 |
| $\beta$ | $\beta$ | 23 | 10 |

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# Project Operator

- Notation:

$$\Pi_{A_1, A_2, \ldots, A_k}(r)$$

where $A_1 \ldots A_k$ are attribute names and $r$ is relation name.

- Result is defined as relation of $k$ columns obtained by erasing columns that are not listed.
- Duplicate rows removed from result, since relations are sets.
  - Example: eliminate *branch_name* attribute of relation *account* (branch_name, account_number, balance)

    $\Pi_{account_number,\ balance}\ (account)$

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# Project Operator – Example

Relation *r*

| A | B | C |
|---|---|---|
| $\alpha$ | 10 | 1 |
| $\alpha$ | 20 | 1 |
| $\beta$ | 30 | 1 |
| $\beta$ | 40 | 2 |

$\Pi_{A,C}(r)$

| A | C |
|---|---|
| $\alpha$ | 1 |
| $\alpha$ | 1 |
| $\beta$ | 1 |
| $\beta$ | 2 |

=

| A | C |
|---|---|
| $\alpha$ | 1 |
| $\beta$ | 1 |
| $\beta$ | 2 |

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# Union Operator

- Notation: $r \cup s$
- Defined as:

  $r \cup s = \{t \mid t \in r \text{ or } t \in s\}$
- For $r \cup s$ to be valid.
  - $r$, $s$ must have *same* **arity** (same number of attributes)
  - Attribute domains must be **compatible** (example: 2nd column of $r$ deals with same type of values as does 2nd column of $s$)
- Example: find names of all customers with either account or loan

  $\Pi_{customer_name}(depositor) \cup \Pi_{customer_name}(borrower)$

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# Union Operator – Example

Relations *r*, *s*:

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\beta$ | 1 |

*r*

| A | B |
|---|---|
| $\alpha$ | 2 |
| $\beta$ | 3 |

*s*

r ∪ s:

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\beta$ | 1 |
| $\beta$ | 3 |

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# Set Difference Operator

- Notation $r - s$
- Defined as:

$$r - s = \{t \mid t \in r \text{ and } t \notin s\}$$

- Set differences must be taken between **compatible** relations.
  - $r$ and $s$ must have **same arity**
  - Attribute domains of $r$ and $s$ must be **compatible**

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# Set Difference Operator – Example

Relations $r$, $s$

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\beta$ | 1 |

$r$

| A | B |
|---|---|
| $\alpha$ | 2 |
| $\beta$ | 3 |

$s$

$r - s$

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\beta$ | 1 |

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# Cartesian-Product Operator

- Notation $r \times s$
- Defined as:

  $r \times s = \{tq \mid t \in r \text{ and } q \in s\}$
- $tq$ is concatenation of $t$ and $q$
  - Assume that attributes of r(R) and s(S) are **disjoint**, that is, $R \cap S = \varnothing$.
  - If attributes of $r(R)$ and $s(S)$ are not disjoint, then **renaming** must be used.

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# Cartesian-Product Operator – Example

Relations $r$, $s$

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\beta$ | 2 |

$r$

| C | D | E |
|---|---|---|
| $\alpha$ | 10 | a |
| $\beta$ | 10 | a |
| $\beta$ | 20 | b |
| $\gamma$ | 10 | b |

$s$

$r \times s$

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |

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# Composition of Operators

- Expressions with multiple operators

$r \times s$

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |

$\sigma_{A=C}(r \times s)$

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |

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# Rename Operator

- Rename can be used to refer to results of relational algebra expressions more than one name.
  - Example:

$$\rho_X(E)$$

returns expression $E$ under name $X$

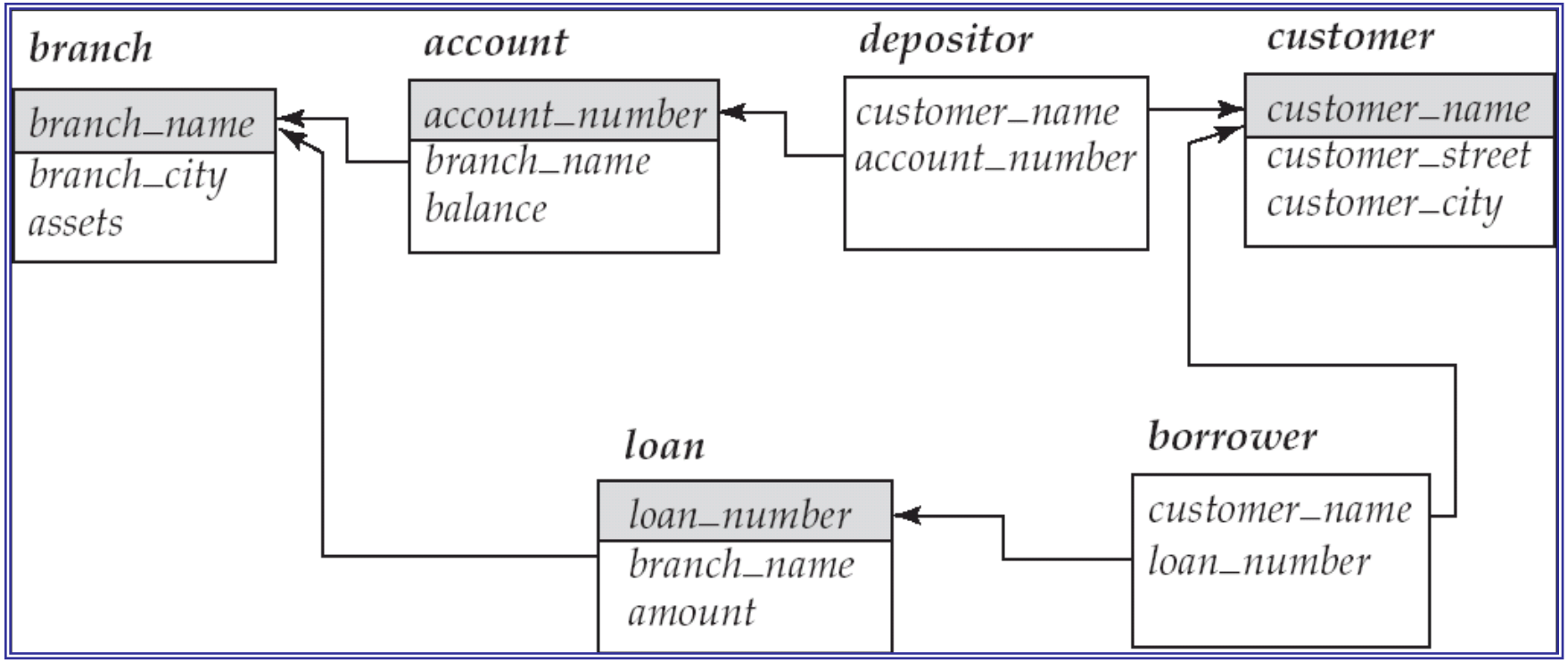
- If relational algebra expression $E$ has arity $n$, then

$$\rho_{X(A_1,A_2,\ldots,A_n)}(E)$$

returns result of expression $E$ under name $X$, and with attributes renamed to $A_1, A_2, \ldots, A_n$.

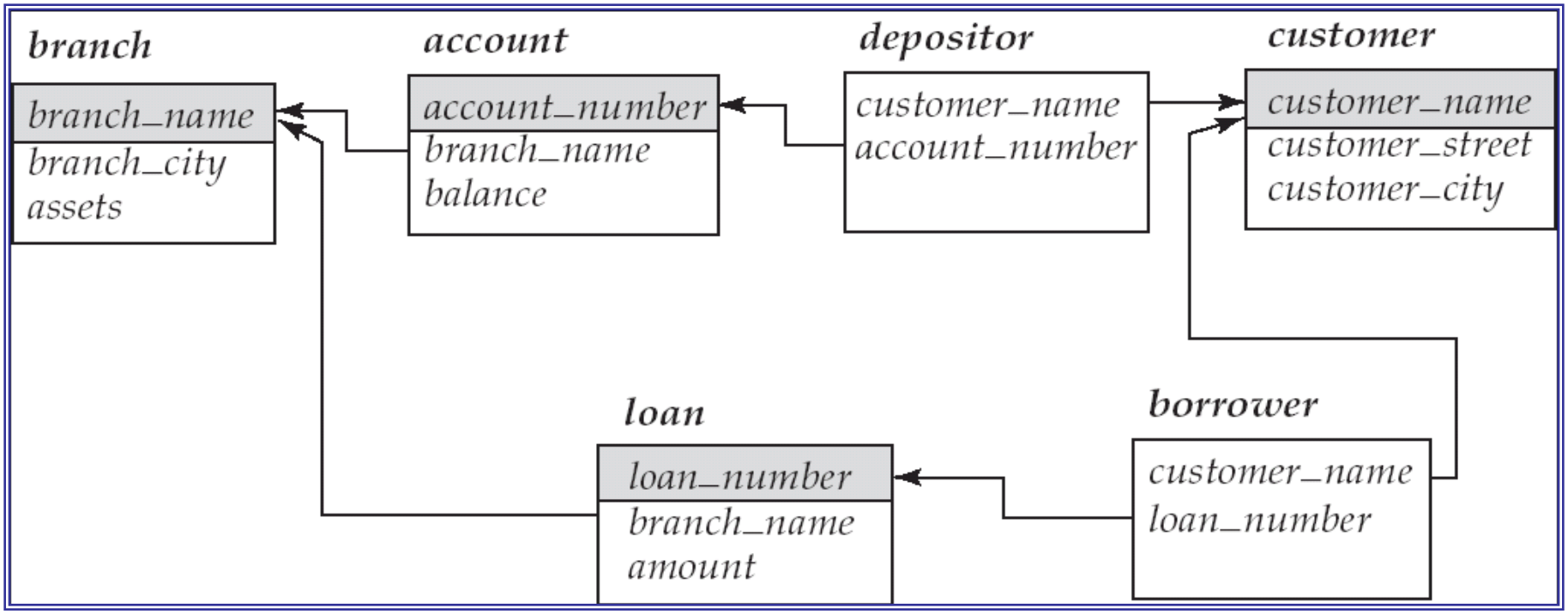
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# Banking Example




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# Example Queries

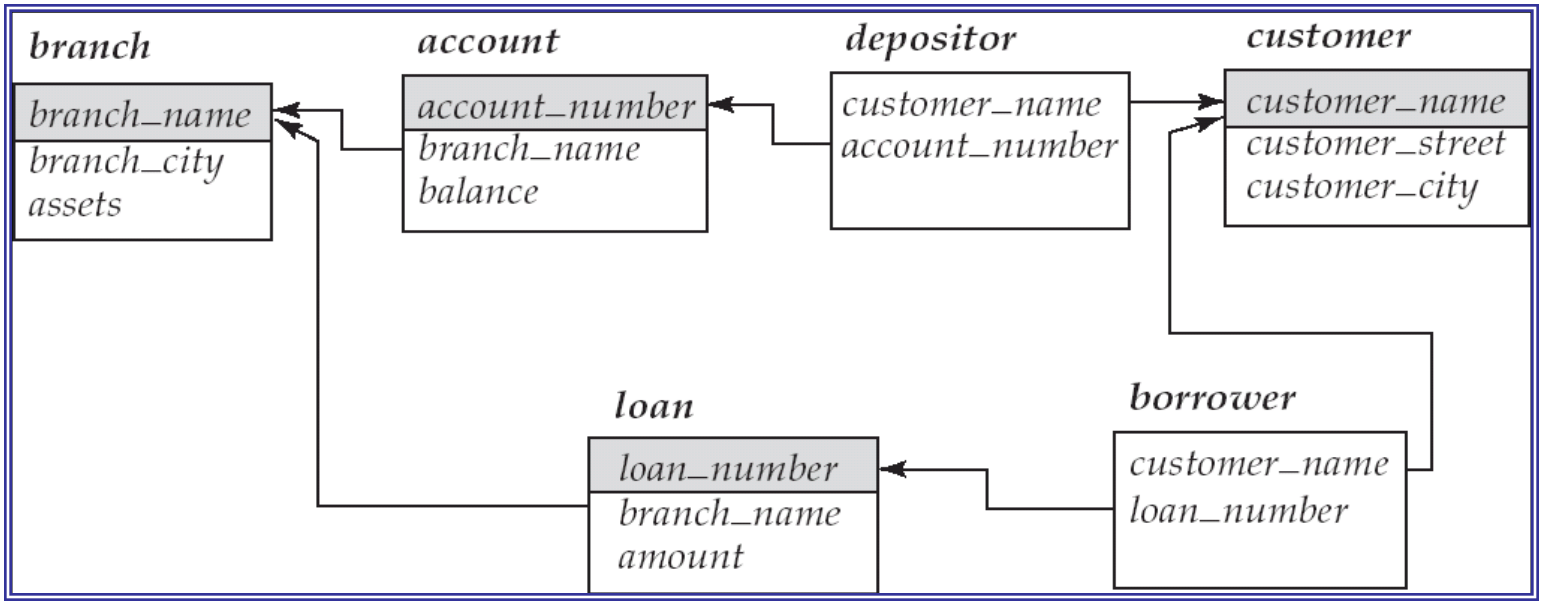



Find all loans of over \$1,200

$$\sigma_{amount > 1{,}200}(loan)$$

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# Example Queries cont'd




- Find names of all customers who have loan at Perryridge branch.

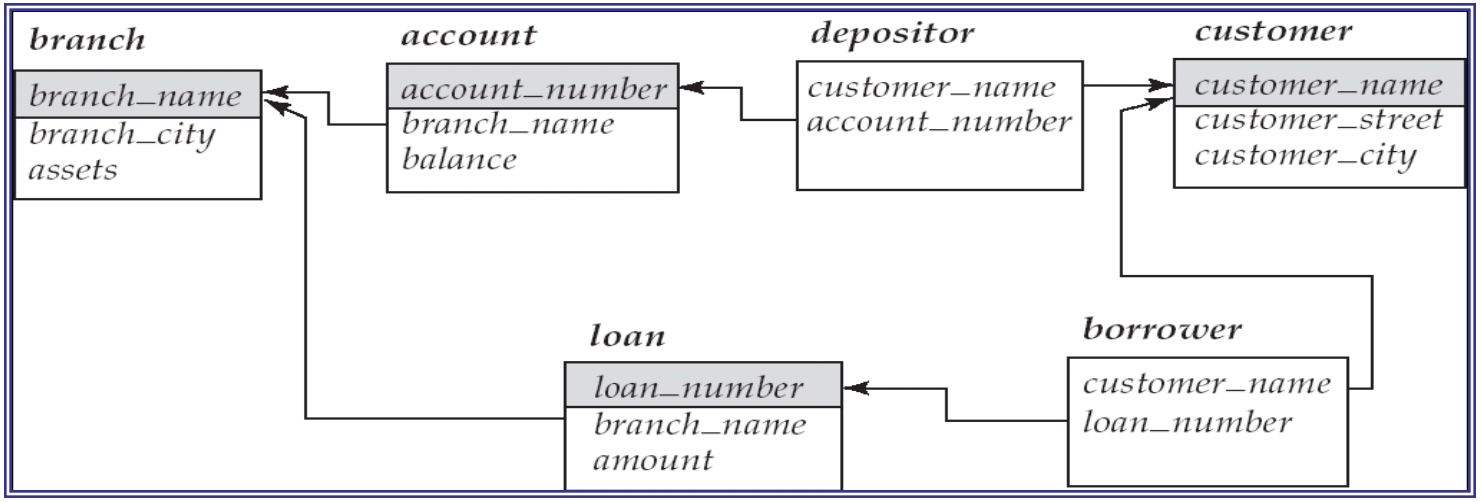
$$\Pi_{\text{customer_name}} (\sigma_{\text{branch_name} = \text{"Perryridge"}} ( \sigma_{\text{borrower.loan_number} = \text{loan.loan_number}} (\text{borrower} \times \text{loan})))$$

OR

$$\Pi_{\text{customer_name}} (\sigma_{\text{loan.loan_number} = \text{borrower.loan_number}} ( (\sigma_{\text{branch_name} = \text{"Perryridge"}} (\text{loan})) \times \text{borrower}))$$

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# Example Queries cont'd




- Find largest account balance
  - Strategy:
    - Find those balances that are *not* largest
      - Rename *account* relation as *d* so that we can compare each account balance with all others
    - Use set difference to find those account balances that were *not* found in earlier step.

$$\Pi_{balance}(account) - \Pi_{account.balance}$$

$$(\sigma_{account.balance < d.balance} (account \times \rho_{d} (account)))$$

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# Additional Operators

- These additional operators do not add any power to relational algebra, but *simplify* common queries.
  - Set intersection: $\cap$
  - Natural join: $\bowtie$
  - Division: $\div$
  - Assignment: $\leftarrow$

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# Set Intersection Operator

- Notation: $r \cap s$
- Defined as:
- $r \cap s = \{ t \mid t \in r \text{ and } t \in s \}$
- Require:
  - $r$, $s$ have *same* ***arity***
  - Attributes of $r$ and $s$ are **compatible**
- Note: $r \cap s = r - (r - s)$

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# Set Intersection Operator – Example

Relation *r*, *s*

| A | B |
|---|---|
| α | 1 |
| α | 2 |
| β | 1 |

*r*

| A | B |
|---|---|
| α | 2 |
| β | 3 |

*s*

$r \cap s$

| A | B |
|---|---|
| α | 2 |

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# Natural Join Operator

- Notation: $r \bowtie s$
- Let $r$ and $s$ be relations on schemas $R$ and $S$, respectively. Then, $r \bowtie s$ is relation on schema $R \cup S$ obtained as follows:
  - Consider each pair of tuples $t_r$ from $r$ and $t_s$ from $s$.
  - If $t_r$ and $t_s$ have same value on each of attributes in $R \cap S$, add tuple $t$ to result, where
    - $t$ has same value as $t_r$ on $r$
    - $t$ has same value as $t_s$ on $s$
- Example:

  $R = (A, B, C, D)$

  $S = (E, B, D)$
  - Result schema = $(A, B, C, D, E)$
  - $r \bowtie s$ is defined as:

$$\Pi_{r.A,\, r.B,\, r.C,\, r.D,\, s.E}(\sigma_{r.B = s.B \wedge r.D = s.D}(r \times s))$$

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# Natural Join Operator – Example

| A | B | C | D |
|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | a |
| $\beta$ | 2 | $\gamma$ | a |
| $\gamma$ | 4 | $\beta$ | b |
| $\alpha$ | 1 | $\gamma$ | a |
| $\delta$ | 2 | $\beta$ | b |

*r*

| B | D | E |
|---|---|---|
| 1 | a | $\alpha$ |
| 3 | a | $\beta$ |
| 1 | a | $\gamma$ |
| 2 | b | $\delta$ |
| 3 | b | $\in$ |

*s*

r ⋈ s

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | a | $\alpha$ |
| $\alpha$ | 1 | $\alpha$ | a | $\gamma$ |
| $\alpha$ | 1 | $\gamma$ | a | $\alpha$ |
| $\alpha$ | 1 | $\gamma$ | a | $\gamma$ |
| $\delta$ | 2 | $\beta$ | b | $\delta$ |

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# Division Operator

- Notation: $r \div s$
  - Often suited to queries that include phrase "**for all**".
- Let $r$ and $s$ be relations on schemas $R$ and $S$, respectively, where
  - $R = (A_1, \ldots, A_m, B_1, \ldots, B_n)$
  - $S = (B_1, \ldots, B_n)$

  Result of $r \div s$ is relation on schema $R - S = (A_1, \ldots, A_m)$

  $$r \div s = \{ t \mid t \in \prod_{R-S}(r) \wedge \forall u \in s ( tu \in r ) \}$$

  Where $tu$ means **concatenation** of tuples $t$ and $u$ to produce single tuple.

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# Division Operator – Example

Relations *r, s*

*r*

| *A* | *B* |
|---|---|
| $\alpha$ | 1 |
| $\alpha$ | 2 |
| $\alpha$ | 3 |
| $\beta$ | 1 |
| $\gamma$ | 1 |
| $\delta$ | 1 |
| $\delta$ | 3 |
| $\delta$ | 4 |
| $\epsilon$ | 6 |
| $\epsilon$ | 1 |
| $\beta$ | 2 |

*s*

| *B* |
|---|
| 1 |
| 2 |

$r \div s$:

| *A* |
|---|
| $\alpha$ |
| $\beta$ |

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# Another Division Example

Relations *r, s*

*r*

| *A* | *B* | *C* | *D* | *E* |
|---|---|---|---|---|
| $\alpha$ | a | $\alpha$ | a | *1* |
| $\alpha$ | a | $\gamma$ | a | *1* |
| $\alpha$ | a | $\gamma$ | b | *1* |
| $\beta$ | a | $\gamma$ | a | *1* |
| $\beta$ | a | $\gamma$ | b | *3* |
| $\gamma$ | a | $\gamma$ | a | *1* |
| $\gamma$ | a | $\gamma$ | b | *1* |
| $\gamma$ | a | $\beta$ | b | *1* |

*s*

| *D* | *E* |
|---|---|
| a | *1* |
| b | *1* |

$r \div s$

| *A* | *B* | *C* |
|---|---|---|
| $\alpha$ | a | $\gamma$ |
| $\gamma$ | a | $\gamma$ |

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# Division Operator Explained

- Division property
  - Let $q = r \div s$
  - $q$ is **largest** relation satisfying $q \times s \subseteq r$
- Definition in terms of basic algebra operation: let $r(R)$ and $s(S)$ be relations, and let $S \subseteq R$

  $$r \div s = \Pi_{R-S}(r) - \Pi_{R-S}((\Pi_{R-S}(r) \times s) - \Pi_{R-S,S}(r))$$
- Explaination
  - $\Pi_{R-S,S}(r)$ simply reorders attributes of $r$
  - $\Pi_{R-S}((\Pi_{R-S}(r) \times s) - \Pi_{R-S,S}(r))$ gives those tuples t in $\Pi_{R-S}(r)$ such that for tuple $u \in s$, $tu \notin r$.

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# Assignment Operator

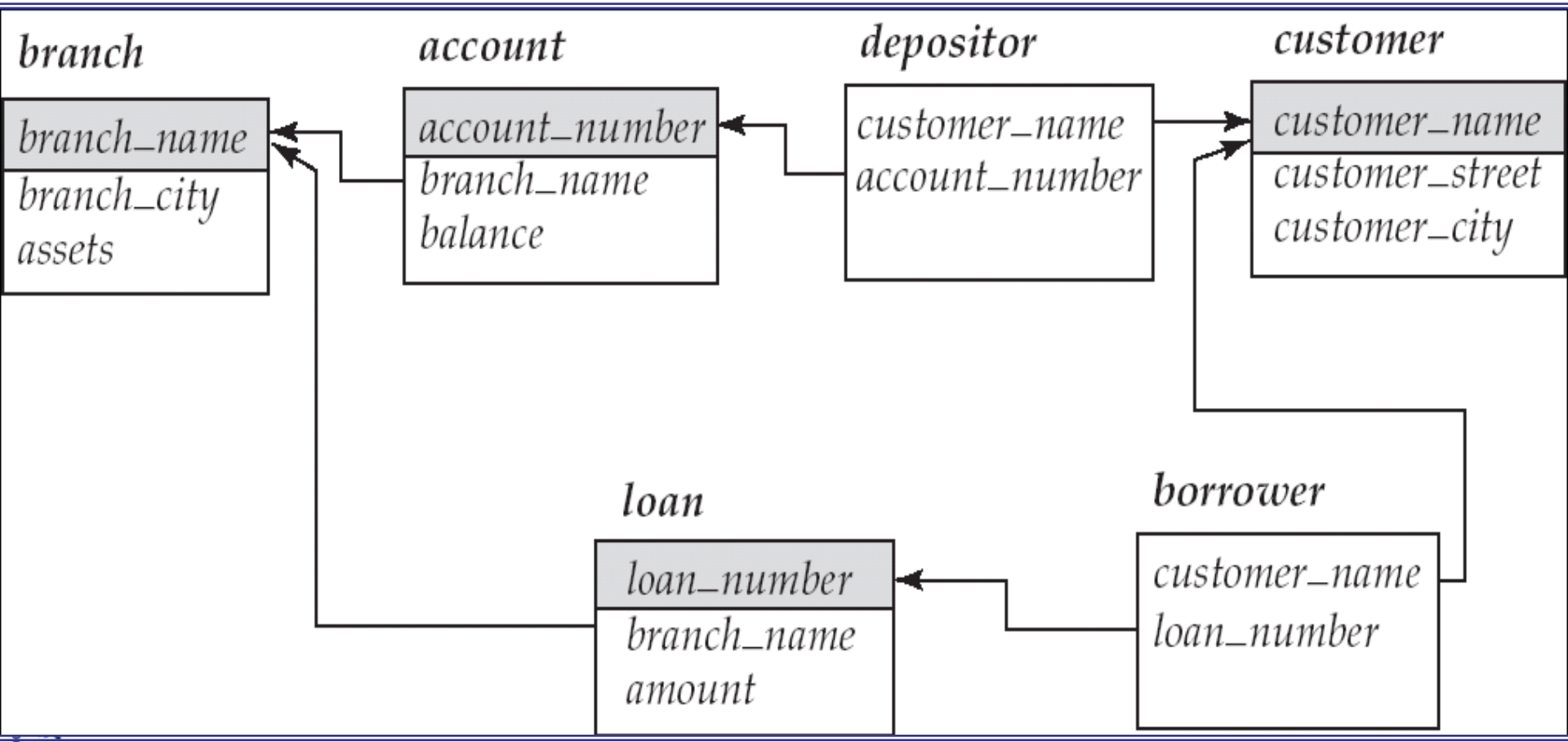
- Assignment operator ($\leftarrow$) provides convenient way to express complex queries.
  - Write query as sequential program consisting of
    - Series of assignments
    - Followed by expression whose value is displayed as result of query.
  - Assignment must always be made to temporary relation variable.
- Example: Write $r \div s$ as

$$temp1 \leftarrow \prod_{R-S}(r)$$
$$temp2 \leftarrow \prod_{R-S}((temp1 \times s) - \prod_{R-S,S}(r))$$
$$result = temp1 - temp2$$

  - Result to right of $\leftarrow$ is assigned to relation variable on left.
  - May use variable in subsequent expressions.

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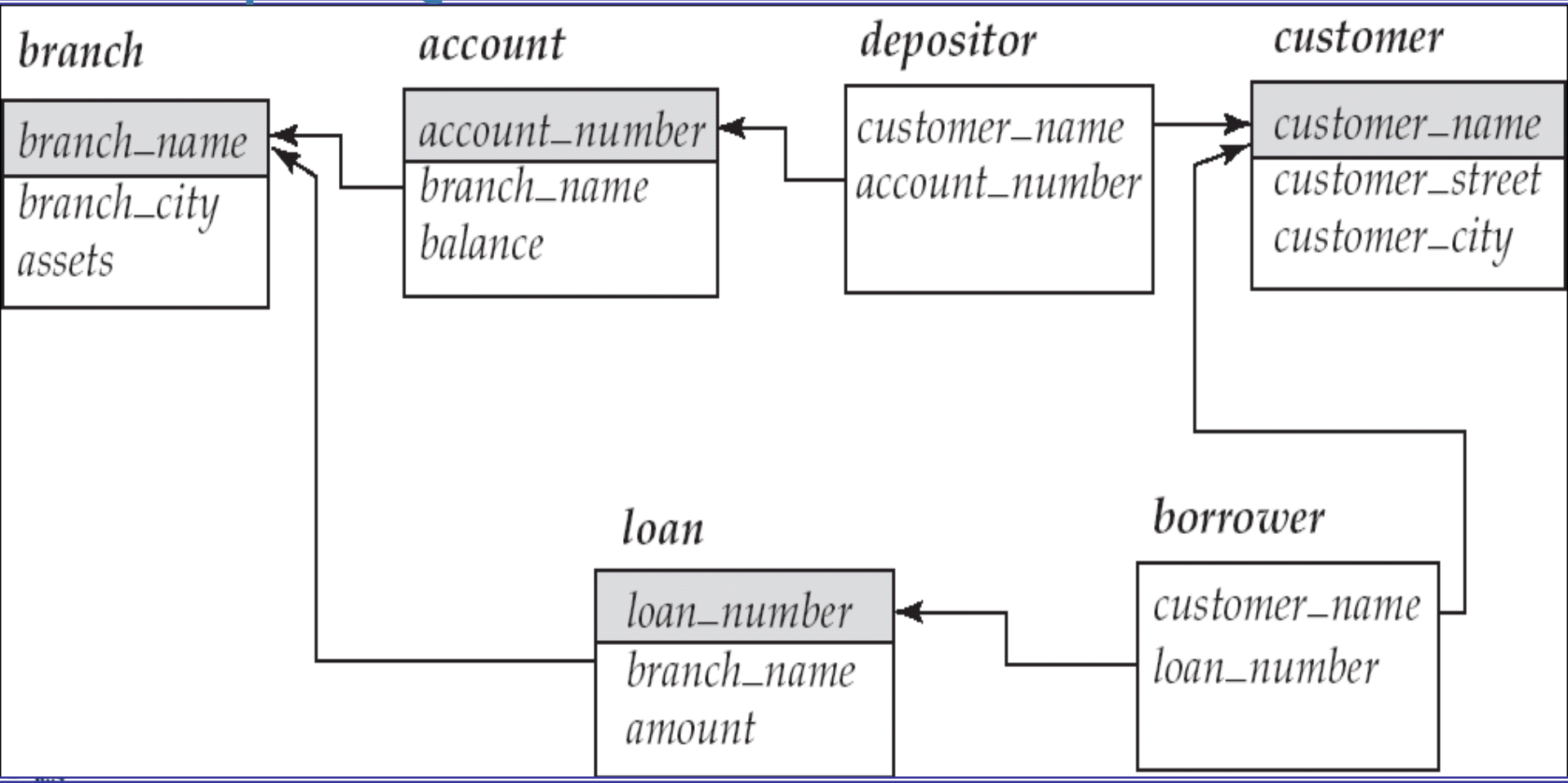
# Example Queries cont'd




Find names of all customers who have both loan and account at bank.

$$\Pi_{customer_name}(borrower) \cap \Pi_{customer_name}(depositor)$$

# Example Queries cont'd

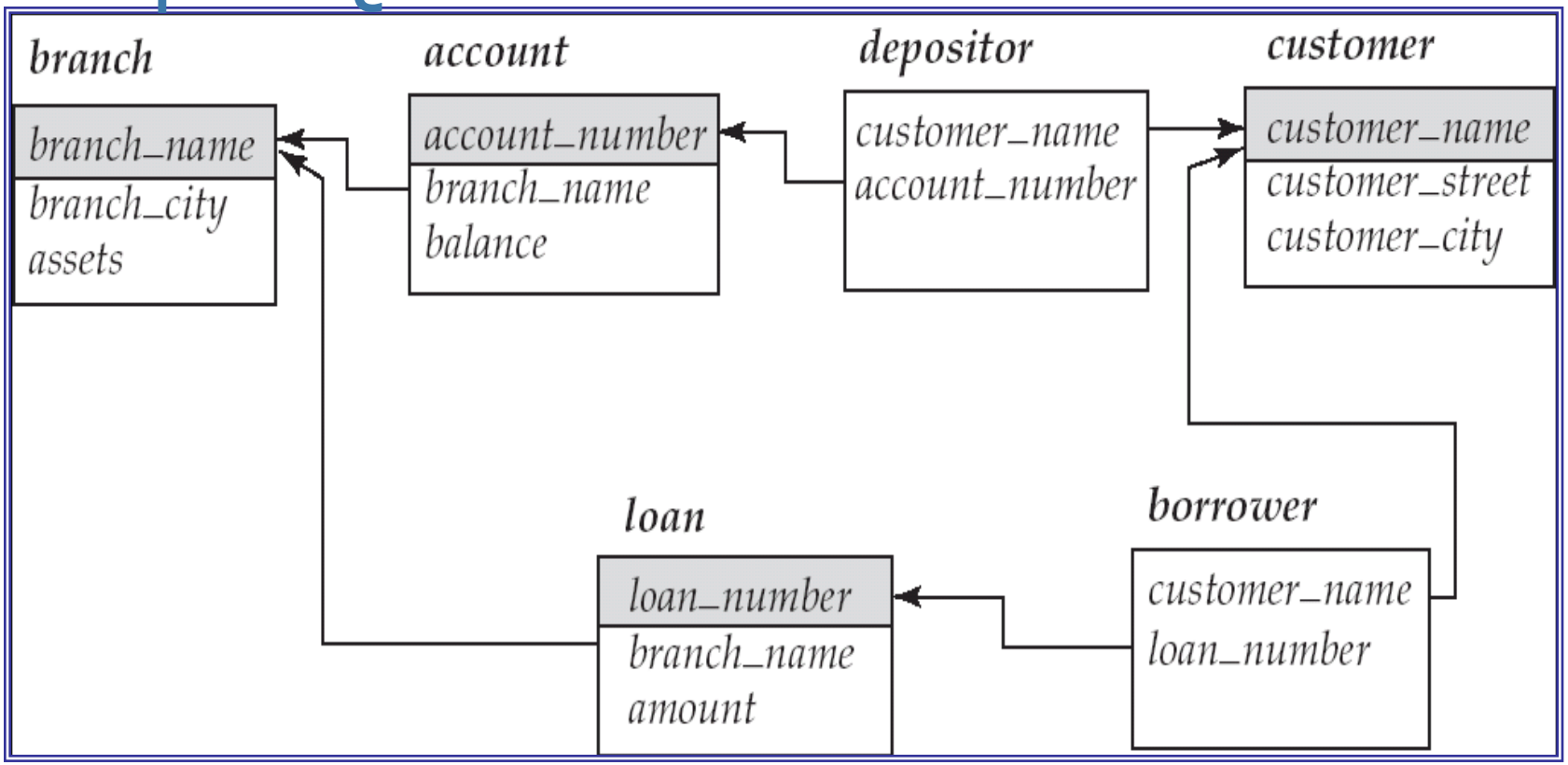



Find name of all customers, their load numbers and loan amount

$$\Pi_{customer_name,\ loan_number,\ amount}\ (borrower \bowtie loan)$$

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# Example Queries cont'd



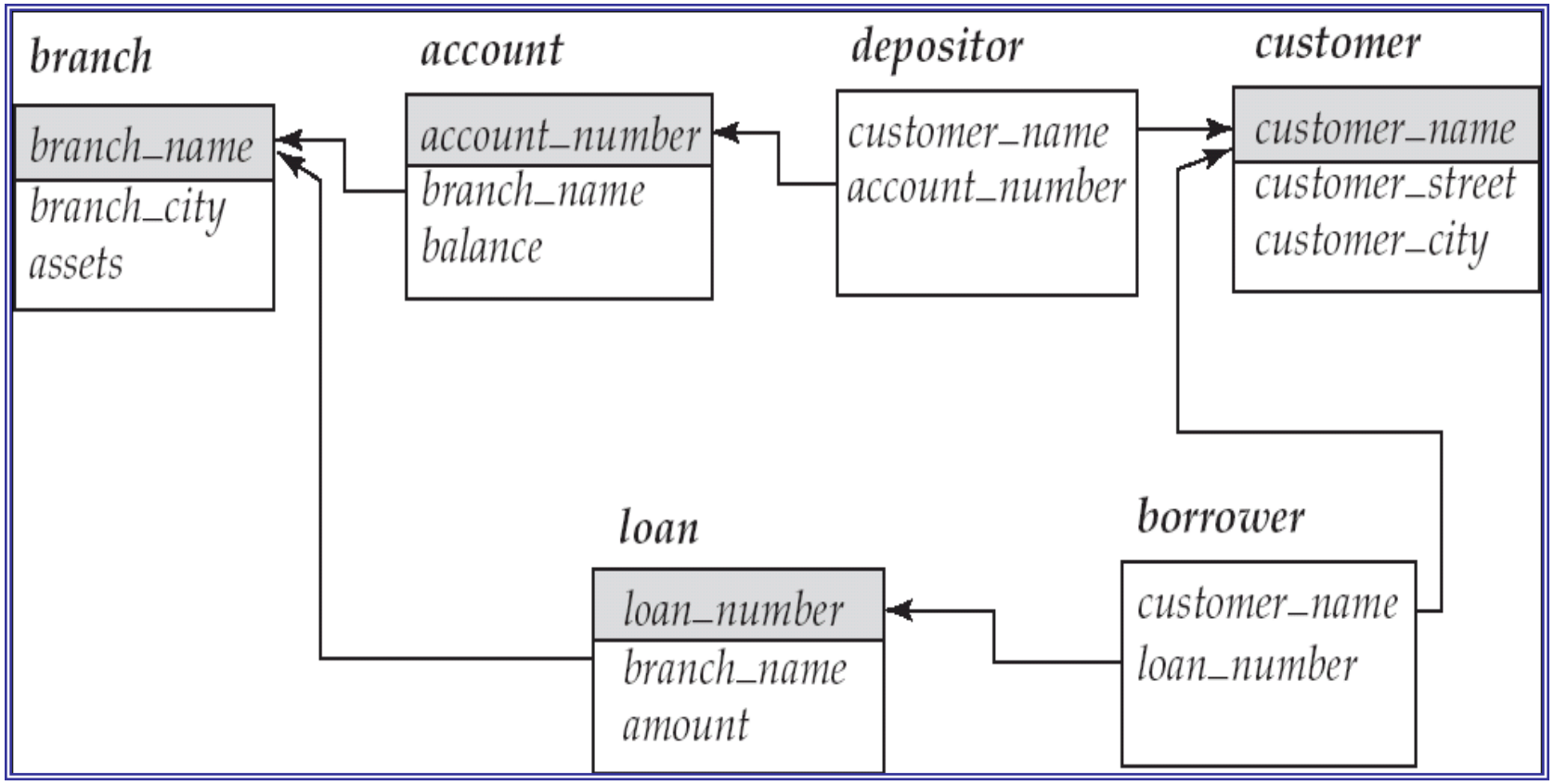

Find names of all customers who have account from both "Downtown" and "Uptown" branches.

$$\Pi_{customer_name}(\sigma_{branch_name = \text{"Downtown"}}(depositor \bowtie account)) \cap$$

$$\Pi_{customer_name}(\sigma_{branch_name = \text{"Uptown"}}(depositor \bowtie account))$$

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# Example Queries cont'd




Find names of all customers who have account at all branches located in Brooklyn city.

$$\Pi_{customer_name,\ branch_name}(depositor \bowtie account) \div \Pi_{branch_name}(\sigma_{branch_city = \text{"Brooklyn"}}(branch))$$

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# End of Lecture

- Reading
  - Textbook 7th edition, chapter 2